

FATIGUE STRENGTH JUSTIFICATION OF COMPOSITE PARTS

Part: **4.9 – COMPOSITE PARTS**
Chapter: **4915**
Title: **FATIGUE STRENGTH JUSTIFICATION OF COMPOSITE PARTS**
Revision: **0.1**
Date: **13/03/2023**

1 INTRODUCTION

This chapter is issued to provide further guidance on how to approach the check of fatigue strength (including quality and non-detectable damage) of structural parts made of composite materials.

As advanced in introductory chapter §4915, common regulations require that composite parts retain UL capability after the whole service life, and the no-damage growth is one of the basic principles of the Fatigue and Damage Tolerance justification of composites. The methodology discussed in this chapter is aligned with this principles, and it is aligned, inspired, and partially copied, from that proposed by Airbus Commercial in Ref. 3. A couple of further remarks:

- This methodology is applicable to Low Cycle Fatigue (LCF) cases, with typical loads spectra not higher than 1E5 cycles
- Only *Level 0* (preliminary) and *Level 1* (simplified) strategies, as named in Ref. 3, shall be discussed, in absence of a significant catalogue of S-N curves for composites in Airbus DS to use in more detailed analyses.

2 METHOD DESCRIPTION

2.1 MATERIALS DESIGN VALUES

Fatigue allowables in composites shall be given in terms of concepts like the *Fatigue Insensitive Threshold* or, in the best case (unlikely for the time being), as S-N curves.

Relevant environmental conditions for these material design values shall typically Room Temperature (RT) with material (partially) moisture saturated in wet ambient at 80-85% HR. On the other hand, material dispersion shall be preferably accounted by knock down factors on the load/stress/strain axis, rather than by scatter factors in the life/cycles axis.

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Fatigue Insensitive Threshold (FIT)

The so called Fatigue Insensitive Threshold, within this scope (not exactly the same definition given by the CMH-17), can be understood as the maximum load/stress/strain of a fatigue cycle which, after being applied to a material/structure up to 1E5 times, will not cause sufficient damage as to degrade significantly the ultimate strength of the part.

Depending on the property/failure-mode, the relevant stress ratio R applicable to this monotonic fatigue cycle could be -1 (eg., for fibrous composite laminates, the plain-strengths, the open-hole and filed-hole strengths, TAI-CAI...) or 0.1 (unfolding, ILSS, bolt bearing or pull-through).

Airbus Commercial present this allowable as a ratio (percentage) of the static strength of the property, as mean value, at RT/Dry conditions. Even that reference point (static strength) is mean value at RT/Dry condition, the ratio points to the correct fatigue design value, i.e. as B-basis and considering moisture absorption (Ref. 3, §3.2.2). Notice that, by using this concept, fatigue margins shall be given as Reserve Factors (as in static strength calculations).

No dedicated tests have been performed up today, within Airbus DS, in order to obtain these values. A first attempt to derive indirectly the Fatigue Insensitive Thresholds from experimental data, can be found in Ref. 2. For some composite materials, Airbus Commercial has released design values for this magnitude. They can be found in Ref. 4.

<i>Property</i>	<i>Fatigue Insensitive Threshold</i>
Strengths driven by inter-laminar stresses, open hole and bolted joints	40% times the ultimate strength
Rest of material strengths (under in-plane loads), including residual strengths after low velocity impact (BVID)	50% times the ultimate strength

Figure 1. Recommended preliminary design values for the Fatigue Insensitive Threshold (as per Ref. 2), not valid for final justification without dedicated experimental validation

Ojo! Los FIT de nuestra NT creo que no se calcularon con la curva S-N en terminus de valores base-B. Comprobar y corregir si es preciso... a ver si aun asi pueden confirmarse esos ratios!

S-N curve

Curve defining the material fatigue performance in terms of cyclic stress (S) against the cycles to failure (N), also known as Wöhler curve. Within this chapter it should be interpreted in a broader manner, with the S a magnitude representing a generic internal load (including stress but also, typically, strain), the most convenient for the design detail of interest.

Not such a standard methodology is actually set for the determination of fatigue allowables of composites from coupons tests. One of the latest approaches followed by Airbus DS is represented by Ref. 6, where ILSS and CAIE properties are to be tested in fatigue:

- Specimens typically tested at RT/Dry (*), RT/AR or RT/W conditions
- Termination criteria: failure, 10% stiffness loss or run-out (at 1E6 cycles)
- Provision of 18 specimens for determining the S-N curve
- Provision of 9 specimens for determining the *Fatigue Insensitive Threshold*

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DEFENCE AND SPACE

Regression analysis of the fatigue tests results, with the life axis in logarithmic scale, is suggested for obtaining the S-N curve (see note (*)). Softwares like NIST RECIPE for the B-basis curve (if not the above-mentioned 0.85 correction factor on the mean values). For the Fatigue Insensitive Threshold, softwares like the CMH-17 STAT-17 can be used to get the B-basis value.

(*) A typical ratio between RT/W and RT/Dry strengths for a fibrous composite material used in aerospace applications, based in an epoxy resin matrix cured at 180°C, is 0.95 (see Ref. 4)

Load Enhancement Factor (LEF) and B-basis correction factor (K_B)

The Loading Enhancement Factor (*LEF*) is the factor applied to nominal fatigue loads in tests in order to cover material dispersion, this approach frequently used with composite parts as alternative to prolonging the test duration by applying a scatter factor in the test cycles (what is usually done in fatigue tests with metallic parts).

Sample Size	One Lifetime Test		1.5 Lifetime Test		Two Lifetime Test	
	A-Basis	B-Basis	A-Basis	B-Basis	A-Basis	B-Basis
1	1.324	1.177	1.291	1.148	1.268	1.127
2	1.308	1.163	1.276	1.134	1.253	1.114
5	1.291	1.148	1.259	1.120	1.237	1.100
10	1.282	1.140	1.250	1.111	1.227	1.091
15	1.277	1.135	1.245	1.107	1.223	1.087
30	1.270	1.130	1.239	1.101	1.217	1.082

Figure 2. CMH-17, Vol. 3, FIGURE 12.6.3.3(d) Typical LEF for $\alpha_R = 20$; $\alpha_L = 1.25$

If a typical *LEF* used in part/component fatigue tests of single specimens is 1.17 (single specimen, B-basis, see previous table), an alternative way of application of this observations would be that the statistical penalization of the fatigue allowables obtained in coupon tests (eg. to the regression curves representing mean values of stress/strain in S-N curves), to get the B-basis allowables, would be using a K_B not lower than 0.85 (1/1.177). Of course, with a sufficient number of specimens, B-basis values could be obtained by statistical analysis of the sample data by using tools as the *STAT-17* (single-point method) or the *NIST RECIPE* (for regression curves).

2.2 LOADS SPECTRA

As proposed by Ref. 3, the fatigue spectra can be simplified in fatigue equivalent loads in order to facilitate the fatigue strength check of composite parts. The fatigue equivalent load is defined as follows: the equivalent stress is such that the application of 1DSG cycle is equivalent in term of damage to the full life fatigue spectrum.

Level 0

In this case, the fatigue equivalent loads are assumed to be equal to the static limit loads

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Level 1

Simplified analysis assuming an envelope of compression and tension loads instead of complete fatigue spectrum. Some examples of envelope loads are:

- ✓ Extreme point of the spectrum
- ✓ Envelope of the calculated fatigue loads: the extreme steady fatigue loads of the mission (mostly the ground-air-ground) and addition of the common and significant perturbations 1 per 1000 flights (gust, dynamic landing,).

2.3 FATIGUE STRENGTH CHECK

When using the simplified methods discussed in previous paragraphs, the fatigue strength check shall be presented, as in static strength calculations, as Reserve Factors, as:

$$RF = \sigma_{fit} / \sigma_{fat}$$

Where σ_{fit} is the fatigue insensitive threshold and σ_{fat} the maximum (or minimum if compression is the relevant component) of the fatigue equivalent loads. The RF should be greater than 1.0.

Note that, if the considered fatigue equivalent loads are those corresponding to *Level 0*, the fatigue RF (RF_{fat}) can be derived from the static RF (RF_{ul}) in a relatively simple calculation:

$$\begin{aligned} RF_{fat} = \sigma_{fit} / \sigma_{fat} &= K_{fit} \cdot \sigma_{all-mean-rt} / \sigma_{II} = K_{fit} \cdot (\sigma_{all-b-hw} / K_B) \cdot (EKDF_{rtw} / EKDF_{hw}) / (\sigma_{ul} / 1.5) = \\ &= 1.5 \cdot K_{fit} \cdot RF_{ul} \cdot EKDF_{rtw} / (K_B \cdot EKDF_{hw}) \end{aligned}$$

Examples

Let's assume the location of interest has a minimum static Reserve Factor of 1.20, is associated to CAI (Compression After Impact)(*) failure, and calculated for a case at HW conditions. Let's assume also that the *Fatigue Insensitive Factor* of that location, including a barely visual impact damage, for cycling loads, is 0.50 times the CAI strength (as mean value) at RT/Dry conditions, i.e.:

- Critical static RF : $RF_{ul} = 1.20$
- EKDF ratio RT/W : $EKDF_{rtw} = 0.95$
- EKDF ratio H/W : $EKDF_{hw} = 0.87$
- B-basis correction : $K_B = 0.83$
- Fatigue Insensitive Threshold : $K_{fit} = 0.50$

Then, the fatigue Reserve Factor, for the *Level 0* strategy, would be:

$$RF_{fat} = 1.5 \cdot K_{fit} \cdot RF_{ul} \cdot EKDF_{rtw} / (K_B \cdot EKDF_{hw}) = 1.5 \cdot 0.50 \cdot 1.20 \cdot 0.95 / (0.83 \cdot 0.87) = 1.18$$

If the same static RF was found in a bolted joint, where the critical failure mode is Filled Hole Tension (or by-pass tension in legacy formulations of the bolted joints method):

- Critical static RF : $RF_{ul} = 1.20$
- EKDF ratio RT/W : $EKDF_{rtw} = 1.00$

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- EKDF ratio H/W : $EKDF_{hw} = 1.00$
- B-basis correction : $K_B = 0.90$
- Fatigue Insensitive Threshold : $K_{fit} = 0.40$

$$RF_{fat} = 1.5 \cdot K_{fit} \cdot RF_{ul} \cdot EKDF_{rtw} / (K_B \cdot EKDF_{hw}) = 1.5 \cdot 0.40 \cdot 1.20 \cdot 1.00 / (0.90 \cdot 1.00) = 0.80$$

... then the *Level 0* strategy would NOT be sufficient to justify the location!

(*) Remark: static failure modes associated to buckling stability are not relevant for this kind of fatigue calculation, as buckling cannot occur at load levels within the fatigue envelope.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
DSG	- / hrs	Design Service Goal, in number of flights or in flight hours
FH	hrs	Flight Hours
FIT / fit	- / MPa	Fatigue Insensitive Threshold
K_B	-	B-basis correction factor, i.e. ratio between a B-basis value (of a mechanical property, typ. a material "strength") and the mean value, if a dataset
LCF		Low Cycle Fatigue (LCF)
LEF	-	Loading Enhancement Factor. In this scope, the factor applied to nominal fatigue loads in tests in order to cover material dispersion, this approach frequently used as alternative to prolonging the test duration by applying a scatter factor in the test cycles (what is usually done in fatigue tests with metallic parts).
LF	-	Life Factor
F&DT		Fatigue and Damage Tolerance
LL		Limit Loads
R	-	Fatigue stress ratio, i.e. is the minimum divided by the maximum stress in a cycle
RF	-	Reserve Factor, ratio between allowable load and applied load
RT/AR	-	Room Temperature, material in As Received conditions (moisture not controlled)
RT/Dry	-	Room Temperature, dry material conditions
RT/W	-	Room Temperature, material moisture saturated in ambient at 70°C and 85%HR
S-N		Curve defining the material fatigue performance in terms of cyclic stress (S) against the cycles to failure (N)
SF	-	Safety Factor, ratio between the Ultimate Loads level and the Limit Loads level, typically equal to 1.5
UL		Ultimate Loads

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REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	CMH-17	2012	COMPOSITE MATERIALS HANDBOOK (VOLUME 3. POLYMER MATRIX COMPOSITES MATERIALS USAGE, DESIGN, AND ANALYSIS)
Ref. 2	TE-T-NT-220018	A	FATIGUE INSENSITIVE THRESHOLD. SUMMARY OF TEST EVIDENCES.
Ref. 3	[Airbus] V029RP0827364	3.0	Composite Fatigue checking Process
Ref. 4	[Airbus] X029RP1439435	11.3	Summary of Design Values for Composite Materials
Ref. 5	[Airbus] AITM1-0075	1.0	Fibre Reinforced Plastics. Constant amplitude fatigue tests on coupons.
Ref. 6	TE-T-SP-220001	B	STRENGTH OF CFRP LAMINATES MADE OF HEXCEL HITAPE/RTM6 UD-TAPE. TEST SPECIFICATION.

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
STAT-17 (CMH-17)	.	A and B-basis calculations for the single-point method
RECIPE (NIST)	.	REgression Confidence Intervals on PErcentiles, performs calculations that determine material basis values from linear models including regression and analysis of variance

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.1 DD/MM/2023	Gustavo Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

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APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Analysis (TEPAA)	José Efraín Mirón Rubio DD/MM/2023	Holger Hickethier DD/MM/2023
Stress Methods (TEPAP-TL4)	Gustavo Baños DD/MM/2023	-

Table 3. Approval signatures table for last revision of this chapter

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